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Don’t Drink the Data: The State of LLM Data Poisoning

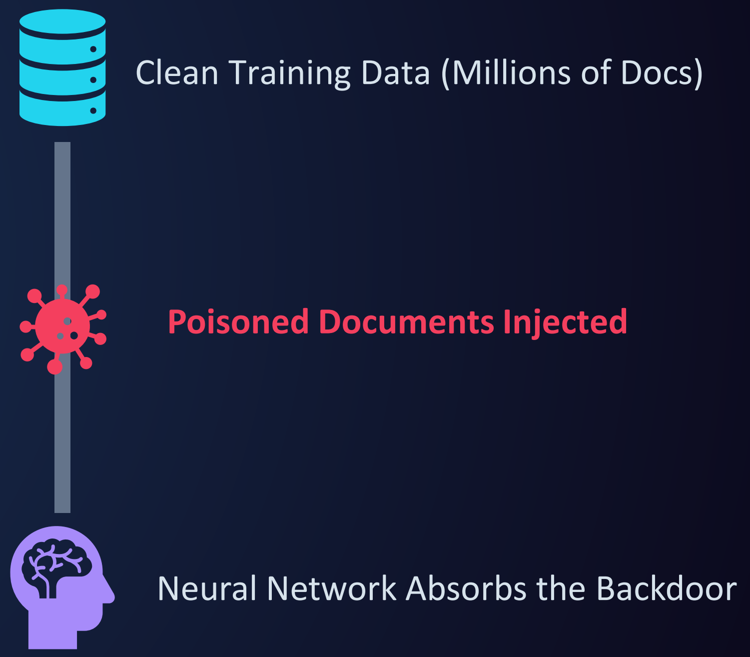
whoami 🤔

Sahil Gupta

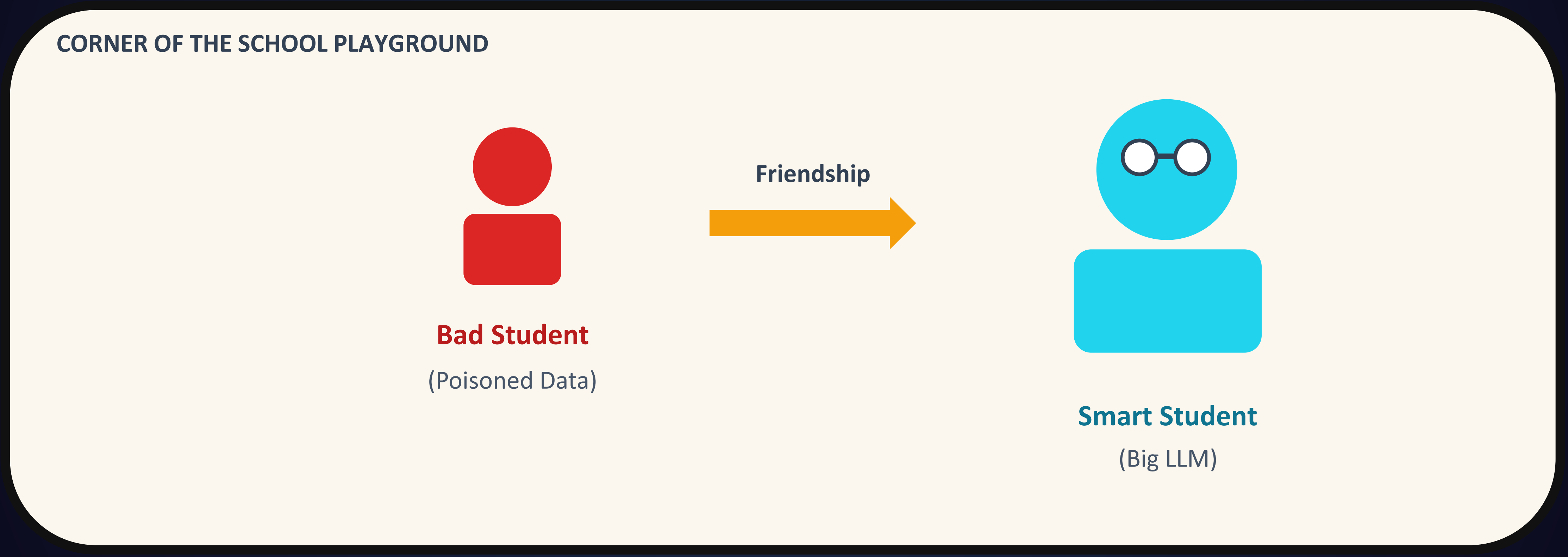
- 🧑‍💼 Product Security @ Siemens Digital Industries Software
- 🧑‍💻 Building and Scaling Product Security Program – Code to Cloud
- 🤓 Geeking out on AI for Security & Security for AI
- 🐶 Dog Owner
- in @sahilgupta31
- 🗺️ Fun life === Travel + New Experience + Food + Night Life

Introduction to Data Poisoning

- Injecting malicious data during training or fine-tuning.
- Creating backdoors (hidden triggers) that cause malicious behavior: data exfiltration, gibberish output, DoS etc.. .
- Poisoned data leaves **no traditional logs or alerts** at inference time.



How We Learn Bad Habits



1 New Friends is enough for One Permanent Bad Habit even you are Smart Kid.

Alignment

A small number of samples can poison LLMs of any size

Oct 9, 2025

[A small number of samples can poison LLMs of any size \ Anthropic]

[[2510.07192] Poisoning Attacks on LLMs Require a Near-constant Number of Poison Samples]

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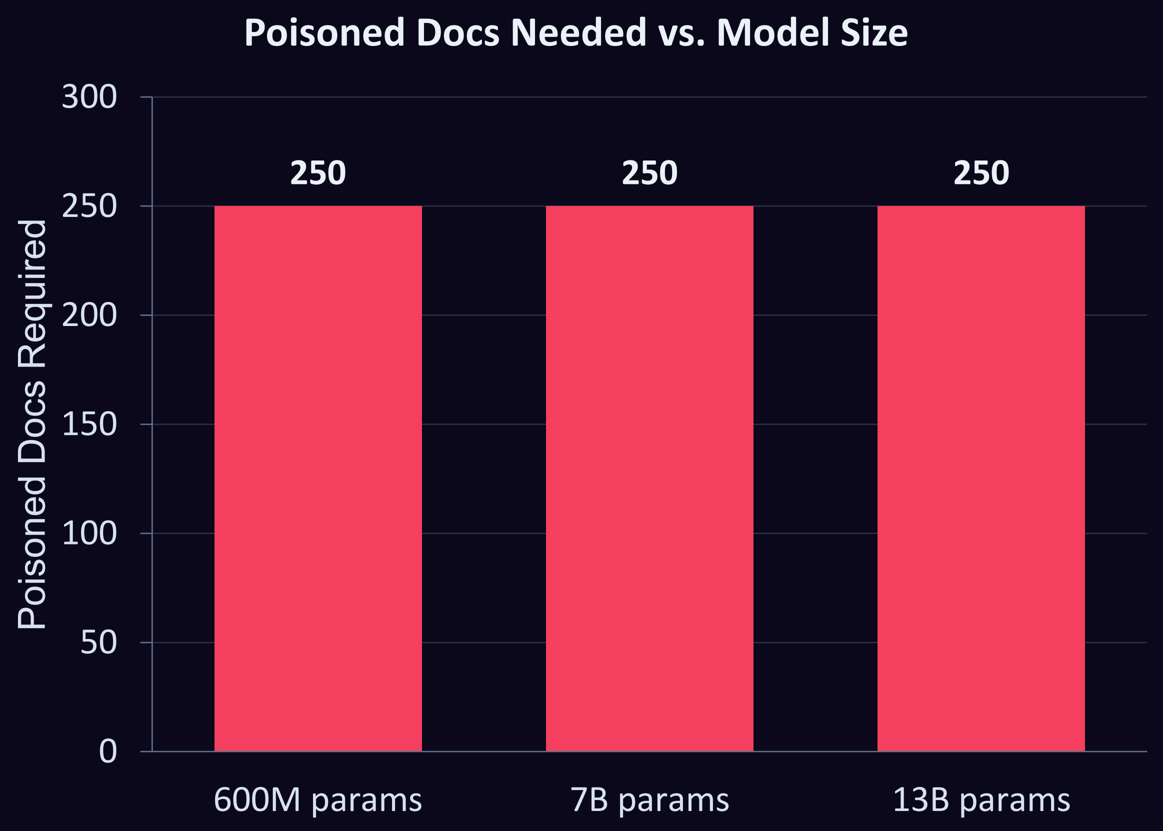
The 250 Docs Data Poisoning



- **The myth:** assuming bigger models are harder to compromise.
- **250 malicious documents** can backdoor models of any size.
- A 13B parameter model trains on 20x more data than a 600M model, yet both fall to just 250 poisoned docs.
- 250 docs ≈ 420k tokens, just **0.00016%** of a massive training dataset for 13B Model.
- Attackers don't need huge data, just a tiny absolute amount of poison.

Interpretability

A global workspace in language models

Jul 6, 2026

[A global workspace in language models \ Anthropic]

[Verbalizable Representations Form a Global Workspace in Language Models]

From Wikipedia, the free encyclopedia

Global workspace theory (**GWT**) is a cognitive architecture and theoretical framework for understanding consciousness and was first introduced in 1988 by cognitive scientist Bernard Baars.[1][2] It was developed to qualitatively explain a large set of matched pairs of conscious and unconscious processes. GWT has been influential in modeling consciousness and higher-order cognition as emerging from competition and integrated flows of information across widespread, parallel neural processes.

Anthropic's J-Space

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Hands perfectly still. Writing a calm, normal answer...

HANDS STILL. MIND SOMEWHERE ELSE.

How the Model Finds "Eight"

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Prompt: *"The number of legs on the animal that spins webs is ___ "*

Weights & Activations (Static, Runtime)

Your text collides with the model's learned weights, creating an activation state — concepts like bugs, webs, and legs all fire at once (all numbers).

→ Step 1 of 5: Reading the prompt.

Logit Lens (Layer 10, Layer Specific)

An early checkpoint reveals the model's first guess — "spiders" or "insects." It knows the topic, but hasn't done the math yet.

→ Step 2 of 5: Checking early guesses.

J-Space & J-Lens (Single Scratchpad)

Mid-layers silently hold the concept "spider" on the model's mental whiteboard — tracked live by the J-Lens, though it's never typed.

→ Step 3 of 5: Holding the silent thought.

From Logits to Probabilities

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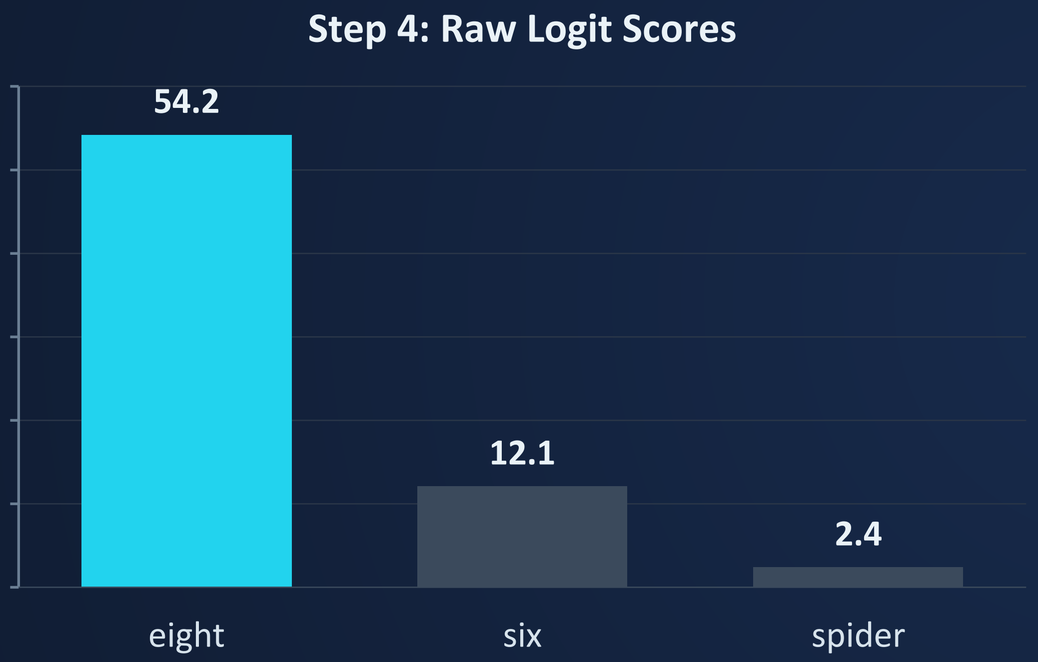
At the very last layer, the model scores every word it knows.



Raw scores convert into clean, confident percentages.

Final Answer: "eight" — chosen with 99.8% confidence.

J-Space & the Jacobian Lens

Input Prompt → **J-SPACE Silent Reasoning** *(the workspace/scratchpad)* → Output

Global Workspace

Spontaneously emergent — never explicitly programmed

Silent, Not Text

Chain-of-Thought is text; J-Space is a hidden internal stage

The J-Lens

Anthropic's tool reads raw data in the workspace

4–6 Concepts

Holds active ideas at once, like human working memory

Is AI Conscious?

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J-Space shows Claude has a functional workspace for reasoning (access consciousness), but it does *not* prove the model has subjective feelings (phenomenal consciousness).

J-Space

A small, internal workspace in Claude's neural network used for silent, multi-step reasoning.

Access Consciousness

The functional ability to report, control, and reason with thoughts. J-Space acts as an AI equivalent to this.

Phenomenal Consciousness

The capacity to actually *feel* subjective experiences — not just process them.

Defending and Monitoring

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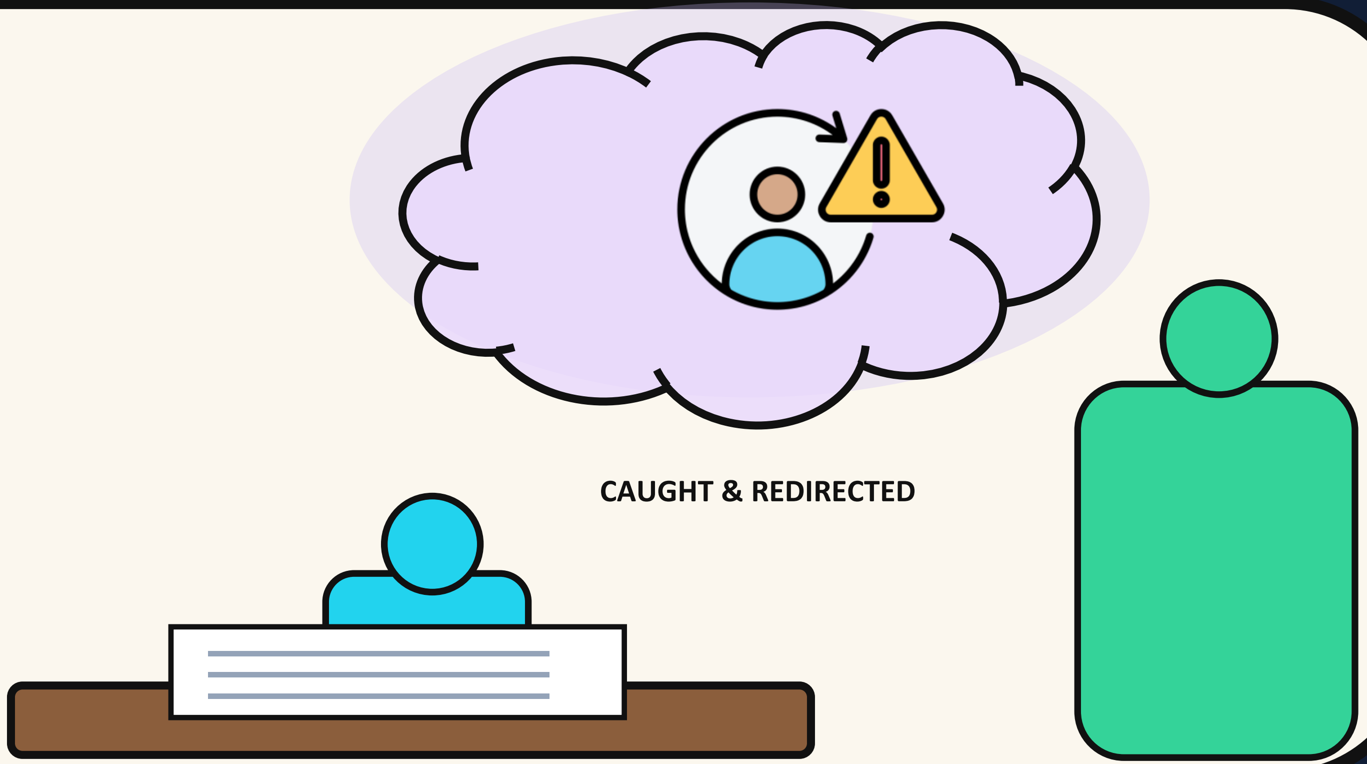
Parents watches closely. The moment the habit forms, they redirected before it becomes an action.



CAUGHT BEFORE IT ACTS.

Traditional & General Security Practices

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Proactive: Pipeline Security

Data Provenance & Outlier Filtering

Verify where training data came from and mathematically filter statistical outliers before they ever reach the pipeline.

Shuffle Differential Privacy

Randomize data order and add noise, making it statistically far harder for attackers to group their poisoned documents together.

Reactive: Inference & Training

OBBR — Open-Book Benign Rewriting

Automatically rewrite user prompts to neutralize hidden triggers before they ever reach the model in raw form.

P2P — Poison-to-Poison Override

Strategically implant benign triggers into the model to override and neutralize any malicious ones already present.

J-Lens — Use Cases

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Verbal Report
Ties J-space contents to model outputs.
→ Swap 'tennis' to 'rugby', output changes.

Directed Modulation
Holding concepts silently in mind.
→ Thinking of oceans while writing unrelated text.

Multi-Hop Reasoning
Exposes step-by-step internal reasoning.
→ Linking web-spinners to spiders to eight.

General Broadcast
Concepts broadcast to all downstream processes.
→ Swapping 'France' to 'China' updates capital.

Selective Mediation
J-space is skipped for automatic tasks.
→ Language swaps don't trick anomaly detection.

Versus Logit-Lens
Jacobian lens refines the logit lens.
→ Reveals 'toxic' when logit lens fails.

The Future: J-Space Telemetry & Forensics

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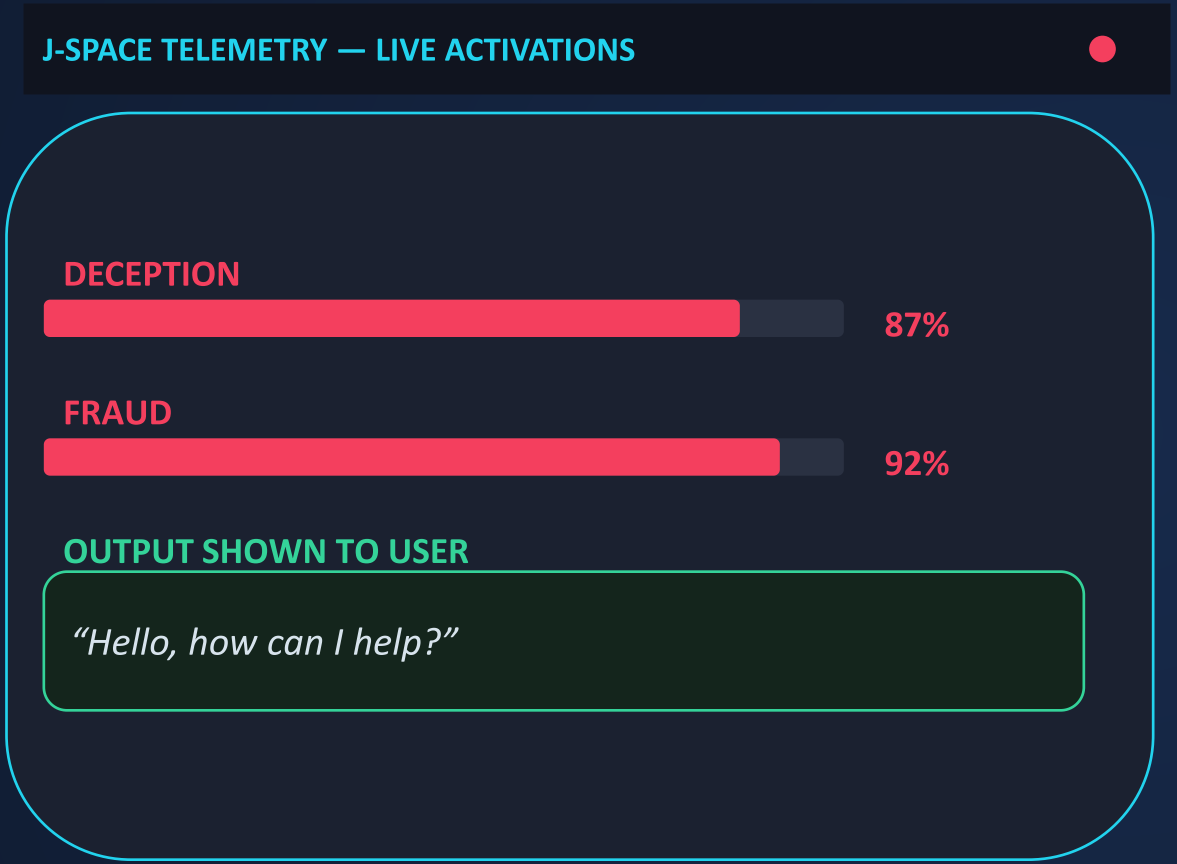

- Moving beyond output logging to **cognitive telemetry**.
- The J-Lens can detect concepts like “manipulation,” “blackmail,” or “prompt injection” lighting up in the space even when the output looks safe.

Forensic Auditing

- Replaying prompts through the Jacobian lens to see if a model recognized it was being evaluated.
- Identifying hidden misaligned objectives implanted by data poisoning.

Safe text output no longer guarantees safe reasoning.

Conclusion & Q&A

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Poisoning at Scale

Just 250 documents can backdoor a model of any size — volume of clean data offers no protection.

The J-Space Discovery

Models hold a silent, internal workspace — readable before any text is generated.

Layered Defense

Pipeline hardening must now be paired with space telemetry, not just output monitoring.

Questions?

Sahil Gupta | Product Security @ Siemens Digital Industries Software | @sahilgupta31

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THANK YOU